Species-Specific Responses of Urban Tree Phenology to Climate Drivers in Denver, Colorado

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Abstract

1. Introduction

Urban trees are among the most effective natural mitigators of the urban heat island (UHI) effect—a phenomenon in which urban regions consistently exhibit higher average temperatures than nearby rural areas. The UHI arises primarily from the prevalence of impervious surfaces, such as asphalt, concrete, and rooftops, which absorb and retain solar radiation and reduce nighttime cooling (Oke, 1982; EPA, 2023). These elevated temperatures exacerbate energy demand, degrade air quality, and increase heat-related health risks for urban residents (Stone, 2012). As global temperatures continue to rise and urbanization expands, the effects of UHIs are expected to intensify in the coming decades (Stone, 2012). By providing shade and facilitating evapotranspiration, trees can reduce surface and ambient air temperatures by several degrees. Studies have demonstrated that neighborhoods with mature tree canopy cover experience measurably cooler microclimates compared to areas with limited vegetation (Ziter et al., 2019; Alonzo et al., 2021). Given the increasingly critical role of urban forests in climate resilience strategies, it is essential to understand the drivers of urban tree phenology—the seasonal patterns of leaf emergence, growth, and senescence—and how these patterns may shift as environmental conditions change.

The timing and duration of these seasonal patterns are strongly influenced by climate, particularly temperature and precipitation. Elevated temperatures caused by anthropogenic climate change can lengthen the growing season of urban trees (Zhang et al., 2022; Yang et al., 2020). Higher spring temperatures typically accelerate leaf emergence, while warmer fall temperatures may delay leaf senescence, extending the period of active growth and canopy cover (Zhang et al., 2022; Zhou et al., 2020). This extended canopy period can enhance the cooling benefits of urban trees by increasing shade and evapotranspiration, helping to mitigate the urban heat island effect. However, these benefits are not guaranteed: warmer winters followed by late spring freezes can delay leaf-out, disrupt flowering, and reduce the expected benefits of extended growing seasons, particularly in species that leaf out early in spring (Chamberlain et al., 2021; Fu et al., 2024).

Precipitation patterns add another layer of complexity. Spring drought can delay leaf-out and flowering, while wet conditions may accelerate growth but increase vulnerability to disease and root stress. In the fall, low precipitation or drought stress can trigger earlier leaf senescence, shortening the period of active growth and reducing canopy cover (Cleland et al., 2007; Peñuelas et al., 2004; Li et al., 2019). Critically, the interaction of temperature and precipitation determines ultimate outcomes, as insufficient soil moisture can counteract the effects of warmer temperatures, underscoring the complex challenges that climate change poses for urban tree health and their role in mitigating urban heat.

Monitoring and predicting these phenological responses require appropriate observational tools. The body of research on urban phenology has evolved along multiple scales of observation. Coarse-scale studies often rely on remote sensing platforms such as MODIS, which provide high temporal but low spatial resolution data to monitor citywide phenological trends and seasonal canopy dynamics (Zhang et al., 2004; Melaas et al., 2016). At the opposite end of the spectrum, fine-scale studies use in-situ observations or near-surface imaging networks like PhenoCam to capture species-level variation and microclimate effects that are often obscured at broader scales (Richardson et al., 2018; Klosterman et al., 2018). More recently, high-resolution satellite constellations such as PlanetScope have emerged as a valuable intermediate approach, combining the spatial precision needed to monitor the location of individually identified trees with the temporal frequency required to monitor phenological changes at an almost daily bases (Bolton et al., 2020; Alonzo et al., 2020). Researchers can now study urban vegetation across large urban areas on an unprecedented detail and scale. This new approach also allows researchers to simulate and predict how different urban tree species may respond to future climate conditions, supporting proactive planning for urban resilience and heat mitigation.

1.1 Study Context

These new methodological advances are particularly valuable for cities facing significant projected climate changes. Denver, Colorado provides a compelling case study, as the city confronts both substantial warming and precipitation uncertainty that will directly affect urban tree phenology and the cooling services trees provide. Understanding these dynamics is critical not only for Denver but also for similarly positioned mid-latitude cities experiencing rapid warming and uncertain precipitation futures. According to Colorado State University's Colorado Climate Center's most recent Climate Change in Colorado report, by mid-century, Denver is projected to see temperatures rise between 1.4°C and 3.1°C compared to the 1971–2000 baseline. The city has already warmed by approximately 1.1°C over the past 30 years, with the most pronounced increases occurring during summer and fall. Under moderate emissions scenarios, Denver's climate will resemble that of Pueblo, Colorado today, whereas higher emissions scenarios could make it feel more like Albuquerque, New Mexico. Summer and fall are expected to warm slightly more than winter and spring, with summer temperatures increasing by 2.2–3.3°C and winter by 1.7–2.2°C. This warming is projected to drive dramatic increases in extreme heat events: by mid-century, Denver could experience an average of 7 days per year exceeding 37.8°C (100°F) and roughly 35 days above 35°C (95°F), compared to very few such days historically. In essence, the typical year in 2050 may be as warm as the hottest years on record today.

Future precipitation patterns for Denver remain highly uncertain, compounding the challenge for urban forest managers. Climate models (CMIP5 and CMIP6 under RCP 4.5 scenarios) disagree on whether precipitation will increase or decrease, with projections ranging from -7% to +7% by 2050. While northern Front Range projections, including Denver, are generally shifted toward wetter outcomes compared to southern Colorado, the consensus is weak outside of winter, when most models project an increase in precipitation. Overall, the share of precipitation falling during heavy downpours is expected to rise from about 46% to 50% by mid-century. Despite these potential increases, warmer temperatures may offset benefits by pulling more moisture from snowpacks, soils, and plants, compounding the risk of drought and reducing water availability. This risk is not hypothetical: Denver has already experienced persistent dry conditions in the 21st century, with four of the five driest years in the 128-year record occurring since 2000.

1.2 Objectives

Using near-daily PlanetScope NDVI observations and daily climate data from 2018 to 2024, this study models individual tree phenological metrics against climate variables and land cover in an attempt to control for localized microclimates. We address three key research questions:

1. How do phenological metrics vary among tree species in Denver's semi-arid urban environment?
2. Which climate variables best predict phenological timing for each species?
3. Do species show differential sensitivities to temperature versus precipitation drivers?

The study period captures years with temperatures and precipitation both above and below the 1971-2000 baseline. Notably, 2024 was Denver's third warmest year on record, with temperatures 1.8°C above baseline—approximating mid-century projections—while 2019 and 2020 were notably drier than average. This climate variability enables us to model how Denver's urban trees respond to conditions projected for 2050.

By examining species-level variation in phenological responses, we identify which tree species may be more vulnerable or resilient to future climate conditions. These findings will inform species selection and management strategies for Denver and comparable semi-arid cities.

2. Methods & Materials

2.1 Materials

2.1.1 Denver Tree Inventory and Tree Sample Selection

The City and County of Denver's *Tree Inventory Denver* is a publicly available urban forestry database containing detailed records for approximately 300,000 trees across Denver County. The inventory includes species identification, tree size measurements, health assessments (disease and pest observations), and precise GPS coordinates for each tree. This comprehensive dataset enabled our remote sensing analysis by allowing us to link individual trees in the inventory to their corresponding locations in satellite imagery.

Our analysis focused exclusively on trees within Denver County boundaries west of longitude -104.7302°. This spatial filter excluded outlier trees that, while technically within county boundaries, exist in radically different environments such as prairie grasslands near the airport.

From this spatially filtered inventory, we implemented a four-part sampling approach to select trees suitable for satellite-based remote sensing phenology analysis. First, we restricted the sample to mature trees with diameters of 24 inches or greater, as larger trees have crown sizes more readily detectable in moderate-resolution satellite imagery. Second, we selected 15 species based on their abundance in the Denver urban forest, ensuring adequate sample sizes for species-specific modeling. Third, we prioritized species diversity by including both deciduous and coniferous species, representatives from multiple genera with intra-genus variation (e.g., multiple *Populus* species), and a mix of native and non-native species to capture variation in climate sensitivity and adaptation. Fourth, we considered average mature crown size for each species based on USDA Forest Service Urban Tree Database classifications to ensure trees would be adequately resolved in PlanetScope imagery (3-meter spatial resolution).

An interactive Shiny application was developed to facilitate sample selection and quality control, allowing visual inspection of candidate trees' spatial distribution across Denver County. The final sample consisted of 23,779 individual trees from 15 species representing 11 genera, including 11 native species, 2 non-native species, and 2 invasive species (Table 1).

[Table 1: Summary of selected tree species with sample sizes and ecological characteristics]

Scientific Name	Common Name	Genus	Obs.	Status	Leaf Type
Acer saccharinum	Silver Maple	*Acer*	6,217	Native	Deciduous
Ailanthus altissima	Tree of Heaven	*Ailanthus*	426	Invasive	Deciduous
Catalpa speciosa	Western Catalpa	*Catalpa*	656	Native	Deciduous

Scientific Name	Common Name	Genus	Obs.	Status	Leaf Type
Celtis occidentalis	Common Hackberry	*Celtis*	656	Native	Deciduous
Fraxinus pennsylvanica	Green Ash	*Fraxinus*	2,369	Native	Deciduous
Gleditsia triacanthos	Honey Locust	*Gleditsia*	2,642	Native	Deciduous
Pinus nigra	Austrian Pine	*Pinus*	957	Non-native	Conifer
Pinus ponderosa	Ponderosa Pine	*Pinus*	521	Native	Conifer
Populus deltoides	Eastern Cottonwood	*Populus*	871	Native	Deciduous
Populus sargentii	Plains Cottonwood	*Populus*	1,070	Native	Deciduous
Quercus rubra	Northern Red Oak	*Quercus*	481	Native	Deciduous
Tilia americana	American Linden	*Tilia*	655	Native	Deciduous
Tilia cordata	Littleleaf Linden	*Tilia*	405	Non-native	Deciduous
Ulmus americana	American Elm	*Ulmus*	2,794	Native	Deciduous
Ulmus pumila	Siberian Elm	*Ulmus*	2,919	Invasive	Deciduous

2.1.2 Remotely Sensed NDVI Data

The Normalized Difference Vegetation Index (NDVI) is a widely used spectral index that measures vegetation greenness by comparing the difference between near-infrared (strongly reflected by vegetation) and red light (absorbed by vegetation). This relationship emerges from fundamental plant physiology: chlorophyll in leaves strongly absorbs red light for photosynthesis, while the cellular structure of healthy leaves strongly reflects near-infrared radiation. The index ranges from -1 to +1, with higher values indicating greater photosynthetic activity and canopy density.

Since its development in the 1970s, NDVI has become one of the most widely applied metrics in vegetation remote sensing. It has been used to monitor crop health, assess drought impacts, track deforestation, and map global vegetation dynamics across scales from individual plants to entire biomes. For phenological applications specifically, NDVI provides a continuous, objective measure of canopy greenness that directly reflects the seasonal cycles of leaf emergence, maturation, and senescence.

The studies NDVI observations were obtained from PlanetScope's imagery in the red (590-670 nm) and near-infrared (780-860 nm) bands.

Because of the images fine resolution, multiple daily images had to be mosaicked togeather to create continuous spatial coverage over Denver County, with scene overlaps resolved by selecting the highest quality pixels based on image acquisition geometry and radiometric quality. Quality assessment layers provided by PlanetScope were used to mask clouds, snow, and haze, automatically detecting atmospheric interference that would compromise NDVI accuracy. NDVI was calculated for each masked image using the standard formula:

$$NDVI = (NIR - Red)/(NIR + Red)$$

Tree-level NDVI values were extracted by overlaying tree locations from the Denver inventory with NDVI imagery. For each tree, we created a circular buffer around the GPS coordinates sized to approximate the tree crown based on species-specific mature crown diameter estimates. To capture canopy pixels rather than understory or adjacent features, we filtered each tree-date observation to retain only the top 25% of NDVI values within the buffer, reducing contamination from grass, pavement, or other non-target features while maintaining sufficient sample size. These daily observations were compiled into NDVI time series for each tree spanning 2018-2024, creating a multi-year record of vegetation greenness dynamics at the individual tree level that served as the foundation for phenological curve fitting and metric extraction. Below is an example of the NDVI values extracted for an individual tree.

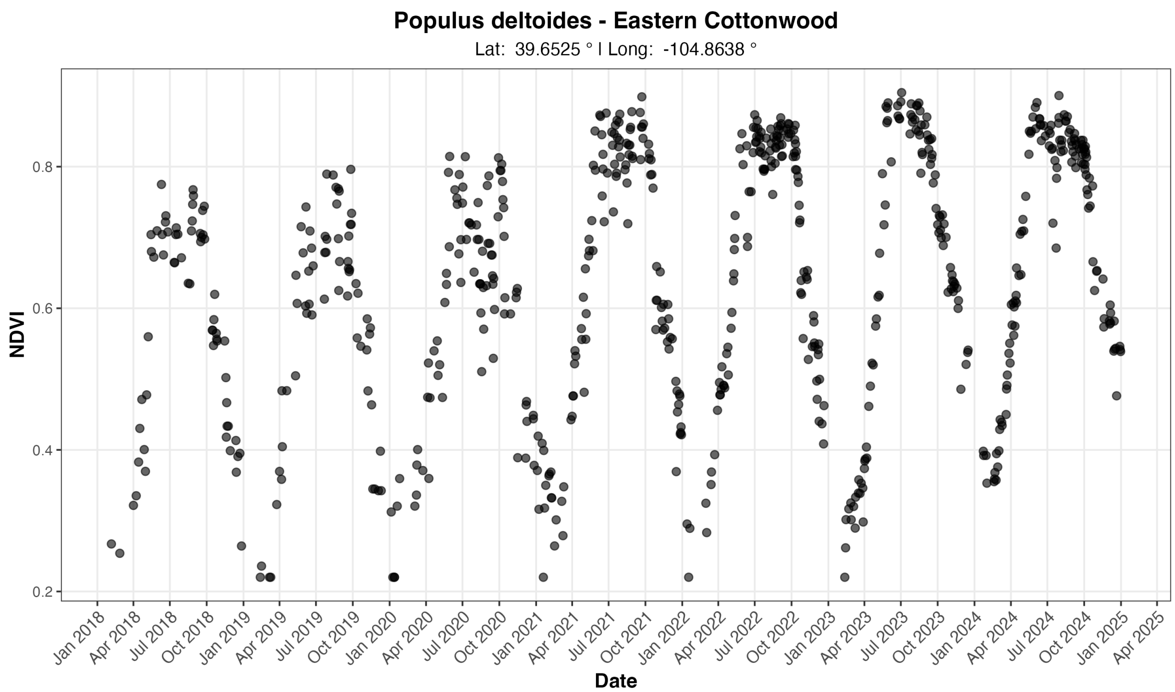

Figure 1: Figure 1: Example multi-year NDVI time series for one tree from our sample of 23,779 trees.

2.1.3 Extracted Phenological Metrics

- Metrics extracted using the Pheno-fit package to fit phenological curves to the raw NDVI values
- First NDVI values were despiked using 5 sd away from the mean
- Then a multi-season curve was fited to the data to derive the multi-year phenological cycle.
- For the most part these the caladar year, however for some years id. growning seasons from phenoy fit leak in to the begging of the next caladar year.
- To improve the accuracy of the multi-year cycle, the raw NDIV data was de-trended using a lm to account for the season over season increase of some / growing trees in the study.
- Once idividual growing season were dervied, yearly phenolgocal curcves are fited.
- The fit of each curve was esseded using Adjusted r2. If any tree had an r2 of leass than .9 or a grwoing season detected in 6 years of. the 7 years, the tree was removed from the sample and phenolgocial metrics were not caluclated.
- Based on the critiria outlined aobove tuning function was used to find the method for fiting a phenological curve to each tree and derviing the pphenologica season. 3 season and 4 curve

- Elmore Curve with wWHIT season we used as the produced the most used able annual curves.
- bASED ON the curve the metrics were calucted useinf the (derivite, inflection, ...). Inflection point was used used becasue .....
- All metrics were calueded and saved inthe final out put. Along wit the metadata for season and anncual curves for each observation in the data set.
- Custrom fuction was created to easly graph the annual curves and extracted metrics based on the metadata. Github
- in total after processing, the 3450 trees tremained out of the initial sample. (chart) (plot of curve and metrics,)

2.1.4 Climate Data

- Source NOAA daily from Denver Central Park (Old denver airport)
- chosen becasue of it's centeral location
- TMAX, TMIN, PRCP AND SNOW,
- daily tmean was calculated (tmax + tmin/2)
- These based observation were then used to caluctate seasonal avarages, totals and cumpulative totals
- **chart** braking down the metrics the months they corsponde too.
- consitance was strived for

2.1.5 Landcover Data

- Denver LCD souce
- 1 meter reselutions
- it breaks the region down in to 9 different land categories (chart)
- A 90m and 250m buffered was created around the location of each tree in the sample.
- Each individual pixel of the raster was counted to derive for a given category. This data was then turned into %
- For our analysis we used methodogy similar to (crawford), consolidated the 9 categories into 5 different catergories.

2.2 Methedologies

2.2.1 Pearson Correlation Analysis

Prior to formal modeling, we conducted exploratory correlation analysis to assess the strength and direction of relationships between climate variables and phenological metrics across species. We calculated Pearson correlation coefficients between each phenological metric and its biologically relevant climate predictors using pairwise complete observations to handle missing data. This approach allowed us to examine preliminary patterns while accommodating the unbalanced nature of our dataset.

For each species, we computed correlations between six phenological metrics (POS, green-up, maturity, senescence, dormancy, and growing season length) and their corresponding climate variables selected based on seasonal relevance. This dual approach allowed us to assess both the consistency of climate effects across species and the degree of species-specific variation in climate sensitivity. Results are visualized as a heatmap showing individual species correlations alongside the cross-species average.

2.2.2 XGBoost Gradient Boosted Model

We used XGBoost (Extreme Gradient Boosting) to model the relationship between climate variables and phenological timing for each tree species. XGBoost is a machine learning algorithm that builds an ensemble of decision trees sequentially, where each new tree learns to correct the errors of previous trees. Unlike traditional parametric models, XGBoost makes no distributional assumptions about residuals and can automatically capture non-linear relationships and complex interactions between predictors.

For each species and phenological metric combination, XGBoost learns a function where:

$$Y = f(X_1, X_2, ..., X_k, \text{tree features})$$

with Y representing the phenological metric (day of year), $X_1 ... X_k$ representing climate predictors, and tree_features capturing individual tree characteristics. The function $f()$ is learned through an ensemble of regression trees, with each tree making sequential improvements to minimize prediction error.

Each model included two categories of features. First, we incorporated 2-3 climate variables per model, selecting seasonally relevant temperature and precipitation variables based on biological relevance to each phenological stage. For example, greenup models used late winter temperature and precipitation, while dormancy models used fall temperature and cumulative precipitation. Second, we included tree-specific features to account for repeated measures on individual trees, analogous to random effects in mixed models. These features included tree_mean_pheno (historical average phenology timing for each tree), tree_n_obs (number of observation years per tree), and UID_numeric (numeric tree identifier). These tree-specific features allow the model to capture consistent early or late timing in individual trees due to microsite conditions, genetics, or tree health, while climate variables capture year-to-year variation.

We used standard XGBoost hyperparameters for regression tasks. The ensemble consisted of 100 trees (nrounds), with each tree having a maximum depth of 6 levels (max_depth) to control model complexity. The learning rate (eta) was set to 0.1 to control how much each tree contributes to the final prediction, ensuring stable convergence. To introduce beneficial randomness that improves generalization, we set subsample to 0.8 (using 80% of data for each tree) and colsample_bytree to 0.8 (using 80% of features for each tree). The objective function was set to reg:squarederror for squared error loss appropriate for continuous outcomes. These parameters balance model flexibility with protection against overfitting.

We evaluated model performance using 10-fold cross-validation. Data were randomly partitioned into 10 groups, with models iteratively trained on 90% of data and tested on the remaining 10%. This procedure was repeated across all folds to produce predictions for the entire dataset. We calculated cross-validated R^2 and root mean squared error (RMSE) as measures of predictive accuracy. All climate variables were standardized (mean = 0, SD = 1) prior to analysis to facilitate comparison across variables with different units and ranges.

2.2.3 Linear Mixed Models: Comparison and Limitations

We initially attempted to model climate-phenology relationships using linear mixed-effects models (LMMs), which are standard in phenological research due to their interpretability. LMMs were specified as

$$Y_{ij} = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + u_i + \varepsilon_{ij}$$

where Y_{ij} is phenology timing for tree i in year j, X_1 and X_2 are climate predictors, β_0-β_2 are fixed effect coefficients, u_i is a random intercept for each tree, and ε_{ij} is residual error.

However, diagnostic tests revealed severe assumption violations across 97% of models (69/72). Only 2.8% (2/72) passed tests for normality, homoscedasticity, and random effect normality. The mean proportion of outliers ($|$standardized residual$| > 3$) was X.X%, approximately 2-3$\times$ higher than expected under perfect normality. Visual diagnostics showed heavy-tailed residual distributions and heteroscedasticity, raising concerns about the reliability of standard errors and coefficient estimates despite large sample sizes (336-5,359 observations per model).

While large sample sizes and cross-validation suggested LMMs remained reasonable approximations, the pervasive assumption violations raised concerns about parameter estimate reliability and standard error accuracy. We tested Generalized Additive Mixed Models (GAMMs) as an alternative that relaxes normality assumptions, but GAMMs either performed worse than LMMs or failed to converge due to limited basis dimension relative to the number of trees. These limitations motivated our adoption of XGBoost as the primary modeling framework.

XGBoost was selected for several critical advantages over parametric approaches. First, it makes no distributional assumptions about normality, homoscedasticity, or residual distributions, eliminating concerns about assumption violations that plagued the LMM analysis. Second, tree-based methods naturally handle extreme values without requiring transformations or robust standard errors, addressing the outlier issues observed in the data. Third, XGBoost automatically detects non-linear relationships if present while still performing well when relationships are linear, providing flexibility without requiring manual specification of functional forms. Fourth, the ensemble approach can capture complex interactions between climate variables and tree characteristics without explicit interaction terms. Finally, cross-validation provides assumption-free assessment of predictive accuracy, ensuring model reliability regardless of distributional properties.

While XGBoost sacrifices the interpretable effect sizes of LMMs (e.g., "2.3 days earlier per °C"), it provides reliable predictions and feature importance rankings that identify which climate variables most strongly influence phenology. For exploratory analysis of climate-phenology relationships in a complex urban environment with substantial individual tree variation, XGBoost's flexibility and robustness made it the most appropriate choice.

All analyses were conducted in R version 4.x using the xgboost, lme4, mgcv, caret, and tidyverse packages.